

Likelihood on the CP-DAG in causal discovery

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Abstract

For a given dataset of observations maximize the likelihood of the causal graph that generates the reconstruction data. Estimate the likelihood of the DAG in the linear causal model. Use Markov equivalence classes of DAGs.

1 Problem statement for graph optimization

There given m observations $\mathbf{x} \in \mathbb{R}^n$ of a random variable X . The structural causal model is

$$\hat{\mathbf{x}} = \mathbf{W}\mathbf{x} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{b}, \boldsymbol{\Sigma}), \quad (1)$$

$\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ is the diagonal noise covariance. Denote by f the model (1) and its parameters $\mathbf{w} = (\mathbf{W}, \mathbf{b})$. The directed acyclic graph G defines the structure of the model f , so that for

$$f_G \stackrel{\text{def}}{=} f(\mathbf{x}; \mathbf{w}, G) : \mathbf{x} \mapsto \hat{\mathbf{x}}$$

a model parameter $w_j = 0$ if the corresponded element of adjacency matrix $\mathbf{G}$ is zero.

The problem is to estimate the most probable DAG structure $p(G \mid X)$ given the observed variable X and the model f :

$$p(G \mid X, f_G) = p(X \mid f_G)p(G) \rightarrow \max_G, \quad (2)$$

where $p(G)$ is a prior over DAGs and $p(X \mid f_G)$ is the likelihood of the data given the DAG structure.

The term $p(X \mid f_G)$ is the marginal likelihood of the data given the DAG structure, obtained by integrating out the parameters $\mathbf{w}$:

$$p(X \mid f_G) = \int p(X \mid \mathbf{w}, f_G)p(\mathbf{w} \mid f_G)d\mathbf{w} \quad (3)$$

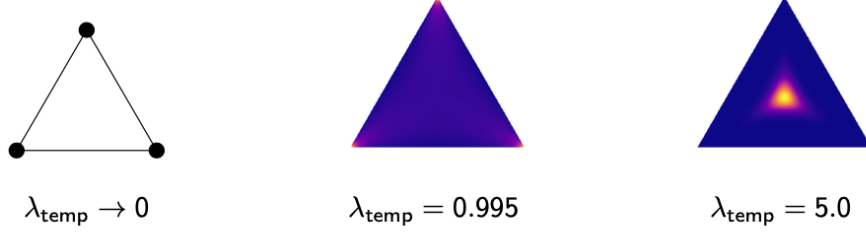
where $p(\mathbf{w} \mid f_G)$ is the prior over the parameters given the DAG structure.

Assume the distribution of the model parameters $p(\mathbf{w} \mid f_G) = \mathcal{N}(\boldsymbol{\mu}, \mathbf{A})$ and with the assumption from (1) rewrite the integrated part from likelihood term (3) as:

$$p(\mathbf{w} \mid \mathbf{x}, f_G) = \frac{1}{2} \sum_i (\mathbf{x}_i - \hat{\mathbf{x}}_i)^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x}_i - \hat{\mathbf{x}}_i) + \frac{1}{2} (\mathbf{w} - \boldsymbol{\mu})^\top \mathbf{A}^{-1} (\mathbf{w} - \boldsymbol{\mu}). \quad (4)$$

The parameters of the prior $\boldsymbol{\mu}, \mathbf{A}$ and covariance matrix $\boldsymbol{\Sigma}$ is estimated during the optimization algorithm. The covariance matrix $\mathbf{A}$ plays its role in the sampling of the nodes of the graph G .

Gumbel-softmax:



Dirichlet

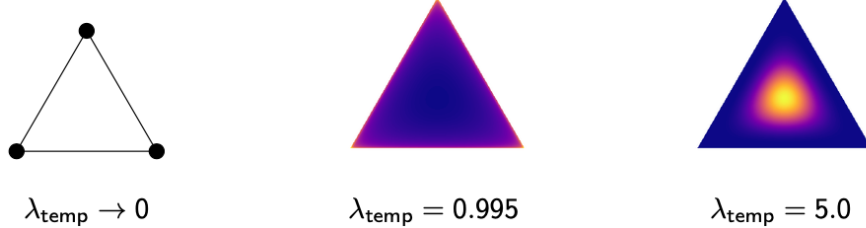


Figure 1: The Dirichlet and Gumbel-Softmax distributions for different values of the temperature τ . As $\tau \rightarrow 0$ the density concentrates near the vertices of the simplex (one-hot vectors). As $\tau = \lambda_{\text{temp}}$ increases, it becomes smoother.

Assume several independent hypotheses of the graph G distribution over the set of DAGs:

- 1) $p(G) = \text{const}$, the uniform prior,
- 2) $p(G) \propto e^{-\gamma|G|}$, a sparsity-inducing prior,
- 3) $p(G) \propto e^{-\gamma h(G)}$, where $h(G)$ is a smooth acyclicity constraint,
- 4) $p(G)$ is a Dirichlet distribution

$$\text{Dir}(\mathbf{y} \mid \boldsymbol{\pi}) = \frac{\Gamma\left(\sum_{i=1}^k \pi_i\right)}{\prod_{i=1}^k \Gamma(\pi_i)} \prod_{i=1}^k y_i^{\pi_i-1},$$

5) $p(G) \propto \text{GS}(\boldsymbol{\pi}, \tau)$, the concrete (relaxed) Gumbel-Softmax distribution

$$p(\mathbf{y} \mid \boldsymbol{\pi}, \tau) = \Gamma(k), \tau^{k-1} \left(\sum_{i=1}^k \frac{\pi_i}{y_i^\tau} \right)^{-k} \prod_{i=1}^k \frac{\pi_i}{y_i^{\tau+1}}.$$

The last two distributions have the parameters $\boldsymbol{\pi} = (\pi_1, \dots, \pi_i > 0, \dots, \pi_k)$, are the unnormalized log-probabilities of edges of the simplex

$$\mathbf{y} \in \Delta^{k-1} = \left\{ \mathbf{y} \in \mathbb{R}^k : y_i > 0, \sum_{i=1}^k y_i = 1 \right\}$$

and τ is the temperature controlling the smoothness of the distribution. Figure 1 illustrates the effect of τ . The Gamma function $\Gamma(k)$ has k categories, edges of the graph G . This is a continuous distribution on the simplex so the change $d\mathbf{y}$ is computable.

State the optimization problem

$$\mathcal{L}(\mathbf{w}; \mathbf{x}, f_G) = \mathcal{L}_{\mathbf{x}} + \mathcal{L}_{\mathbf{w}} + \mathcal{L}_G \rightarrow \min_{G \in \mathcal{G}} \quad (5)$$

using (4) and the listed above prior over DAGs. The problem (5) includes the bayesian hyperparameters:

the diagonal covariance matrix $\boldsymbol{\Sigma}$ controls the noise level in the data,

the mean and covariance $\boldsymbol{\mu}, \mathbf{A}$ control the prior over parameters $\mathbf{w}$,

$\boldsymbol{\pi}$ controls the unnormalized log-probabilities of edges of the graph,

γ controls the sparsity of the graph,

τ controls the smoothness of the distribution over graphs.

The plate notation of the probabilistic model is shown in Figure 2.

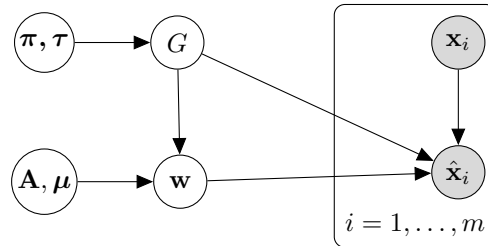


Figure 2: The plate notation of the causal structure model. The nodes $\mathbf{x}_i$ and $\hat{\mathbf{x}}_i$ are observed and reconstructed variables. The parameters $\boldsymbol{\pi}, \tau$ control the distribution over DAGs. The hyperparameters $\mathbf{A}, \boldsymbol{\mu}$ control the prior over the model parameters $\mathbf{w}$.

2 MCMC sampling of the DAG structure

The generic sampling procedure for the DAG structure includes the estimation of the parameters and hyperparameter of the structure causal model.

1. Sample the DAG structure G from the prior distribution $p(G)$.
2. Check its admissibility and the acyclicity constraint $h(G) = 0$.
3. Construct the corresponding structure causal model f_G .
4. Use the prior parameters $\mathbf{A}, \boldsymbol{\mu}$.
5. Optimize the model parameters $\mathbf{w}$ by minimizing the loss function $\mathcal{L}_x$.
6. Sample subsets of the data $\mathbf{x}$ to estimate new hyperparameters $\boldsymbol{\Sigma}, \mathbf{A}, \boldsymbol{\mu}$.
7. Compute the likelihood $\mathcal{L} = \mathcal{L}_{\mathbf{x}} + \mathcal{L}_{\mathbf{w}} + \mathcal{L}_G$.
8. Correct the distribution of the DAG structure G (by accepting or rejecting it based on the likelihood $\mathcal{L}$).
9. Repeat the procedure until convergence of loss $\mathcal{L}$.

Genetic sampling algorithm

The Genetic algorithm as a simple heuristical replacement of the MCMC sampling. Introduce the indicator vector $\mathbf{a} \mapsto G$ that restores an admissible graph structure. This vector is a vertex in the binary cube $\mathbf{a} \in \{0, 1\}^d$, where d is the number of possible edges in the graph. For a graph with n nodes $d \leq \frac{1}{2}n(n-1)$. The genetic algorithm includes the following steps:

1. There is a set of binary vectors

$$\{\mathbf{a}_1, \dots, \mathbf{a}_P\}, \mathbf{a} \in \{0, 1\}^d.$$

2. Get two uniformly random vectors $\mathbf{a}_p, \mathbf{a}_q$, their indexes $p, q \in \{1, \dots, P\}$.
3. Choose a random number $v \in \{1, \dots, n-1\}$.
4. Split both vectors and change their parts:

$$[a_{p,1}, \dots, a_{p,v} \mid a_{q,v+1}, \dots, a_{q,n}] \rightarrow \mathbf{a}'_p,$$

$$[a_{q,1}, \dots, a_{q,v} \mid a_{p,v+1}, \dots, a_{p,n}] \rightarrow \mathbf{a}'_q.$$

5. Choose random numbers $r_1, \dots, r_Q \in \{1, \dots, n\}$.

6. Invert positions $r_1, \dots, r_Q$ of the vectors $\mathbf{a}'_p, \mathbf{a}'_q$.
7. Repeat $P/2$ times the items 2–6.
8. Evaluate the obtained models, compute the loss $\mathcal{L}$.
9. Keep the best P vectors $\mathbf{a}$ with the lowest loss $\mathcal{L}$ and discard the rest.

Repeat the algorithm until convergence of the loss $\mathcal{L}$ or a fixed number of iterations. Here P, Q are the heuristic parameters of the algorithm.

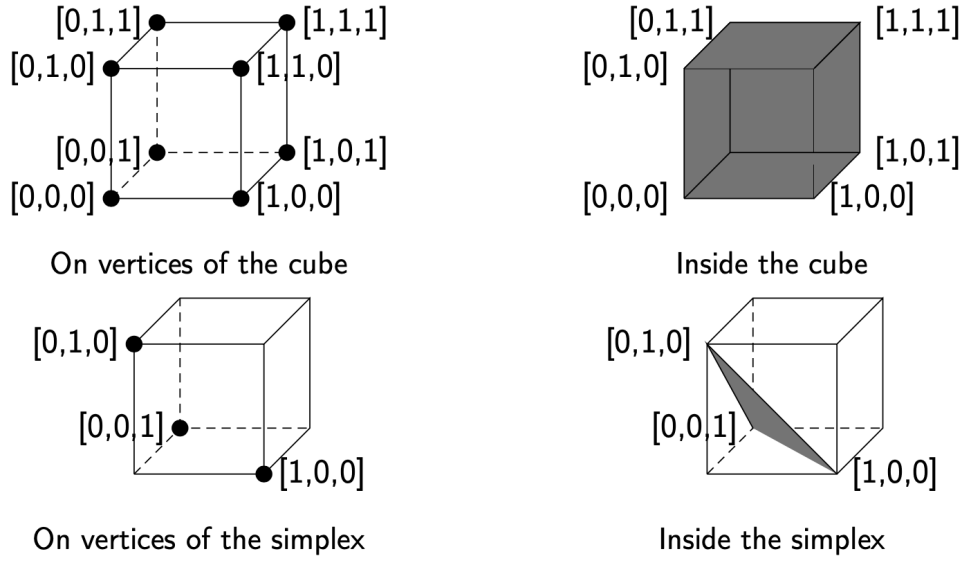


Figure 3: The restrictions to the structure parameters. Examples for some structure parameter.

MCMC sampling algorithm

To sample graphs $\mathbf{a}_0 = \{\mathbf{a} \mid \mathbf{a} \sim p(\mathbf{a} \mid \boldsymbol{\pi})\}$ that deliver models with the best likelihood, use Metropolis-Hastings sampling version of the Markov Chain Monte Carlo sampling algorithm. The basic idea of the algorithm is to generate a sample such that the sample forms a Markov chain. Each element $\mathbf{a}_{t+1}$ of the sample depends on the previous element $\mathbf{a}_t$ of the sample only. Denote by $Q(\mathbf{a} \mid \mathbf{a}')$ an auxiliary distribution $Q(\mathbf{a} \mid \mathbf{a}')$. Choose an initial element $\mathbf{a}_0$ and assign $\mathbf{a}_0 = \{\mathbf{a}_0\}$. Then let an element $\mathbf{a}_t$ be chosen according to the distribution $Q(\mathbf{a}' \mid \mathbf{a}_t)$. The next element $\mathbf{a}'$ is generated randomly. Then the algorithm computes the acceptance ratio a :

$$a = \min_{\mathbf{a}' \in \mathbb{R}^n} \left(\frac{p(D \mid \mathbf{a}', B_0)}{p(D \mid \mathbf{a}_t, B_0)} \frac{Q(\mathbf{a}_t \mid \mathbf{a}')}{Q(\mathbf{a}' \mid \mathbf{a}_t)}, 1 \right).$$

The algorithm accepts the candidate $\mathbf{a}'$ with the probability a ,

$$\mathbf{a}_{t+1} = \mathbf{a}', \quad , W_0 := W_0 \cup \mathbf{a}'.$$

Otherwise, the algorithm rejects the candidate and puts $\mathbf{a}_{t+1} = \mathbf{a}_t$,

$$\mathbf{a}_{t+1} = \begin{cases} \mathbf{a}', & \text{with the probability } a, \\ \mathbf{a}_t, & \text{with the probability } (1 - a). \end{cases}$$

Let the auxiliary distribution $Q(\mathbf{a} \mid \mathbf{a}')$ be normal (it is pretty simplified case; replace to uniform, Dirichlet or Gumbel-Softmax distribution):

$$Q(\mathbf{a} \mid \mathbf{a}') = Q(\mathbf{a}' \mid \mathbf{a}) = \frac{1}{(2\pi\alpha^{-1})^{n/2}} \exp\left(-\frac{\alpha}{2}(\mathbf{a} - \mathbf{a}')^\top(\mathbf{a} - \mathbf{a}')\right).$$

That is, the function $Q(\mathbf{a} \mid \mathbf{a}')$ is symmetric and the acceptance ratio a simplifies to

$$a = \min_{\mathbf{a}' \in \mathbb{R}^n} \left(\frac{p(D \mid \mathbf{a}', B_0)}{p(D \mid \mathbf{a}_t, B_0)}, 1 \right).$$

The initial element $\mathbf{a}_0$ is chosen randomly from the distribution $p(\mathbf{a} \mid \boldsymbol{\pi})$.

Direct optimization of the CP-DAG structure

There are two ways control the model superposition and the DAG structure: 1) direct intervention into the model structure (for example assert a linear model parameter to 0, or apply identical transformation), and 2) control the model structure via the loss function (ridge or lasso regularization).

The problem of continuous optimization of the CP-DAG structure includes:

1. An auxiliary linear space $\mathbf{a}$ that continuously controls the model structure via

$$\frac{\partial \mathcal{L}}{\partial \mathbf{a}} = \frac{\partial \mathcal{L}}{\partial \mathbf{x}} \cdot \frac{\partial \mathbf{x}}{\partial \mathbf{a}}.$$

2. The mapping to the CP-DAG structure G that restores an admissible graph structure

$$\mathbf{a} \mapsto G|_{\text{CP}} \mapsto f_G(\mathbf{w}) : \mathbf{x} \mapsto \hat{\mathbf{x}}.$$

3. Since the model has its parameters $\mathbf{w}$, the optimization of the structure $\mathbf{a}$ is performed along with the optimization of the model parameters $\mathbf{w}$.
4. The optimized parameters $\mathbf{w}$ are used to estimate the bayesian hyperparameters $\mathbf{A}, \boldsymbol{\Sigma}$.
5. The analysis of the hyperparameters corrects the model structure

$$\mathbf{A}, \boldsymbol{\Sigma} \longrightarrow G.$$

The continuous optimization of the DAG structure requires two-step procedure: simple gradient descent step (6) for the structure parameters $\mathbf{a}$

$$\mathbf{a}_{t+1} = \mathbf{a}_t - \eta^2 \frac{1}{|B_t|} \sum_{(x_i) \in B_t} \nabla_{\mathbf{a}} \mathcal{L}(f_G(\mathbf{a}_t; \mathbf{w}, \mathbf{x}_i), \mathbf{\Gamma}, \mathbf{A}(\mathbf{a}_t), \mathbf{\Sigma}) \quad (6)$$

and the stochastic gradient descent (7) step for the model parameters $\mathbf{w}$,

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta^2 \frac{1}{|B_t|} \sum_{(\mathbf{x}_i) \in B_t} \nabla_{\mathbf{w}} \mathcal{L}(\mathbf{w}_t; x_i) \quad (7)$$

with estimation of the Bayesian hyperparameters

$$\mathbf{A}, \mathbf{\Sigma} \text{ estimated with } \mathcal{L}(\mathbf{w}) \text{ by cross-validation.} \quad (8)$$

The algorithm of the continuous optimization of the DAG structure:

1. There given the observations $\{\mathbf{x}_i\} = \bigcup_t B_t$, and the initial DAG G and the model f_G (possible excessive).
2. With roughly optimized parameters $\mathbf{w}$, estimate the initial Bayesian hyperparameters $\mathbf{A}, \mathbf{\Sigma}$.
3. Make (extreme) SVD of the covariance matrix and construct the space for $\mathbf{a}$.
4. Make a gradient step to optimize the graph structure

$$\nabla_{\mathbf{a}} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}}.$$

5. Update or **sample** (perturbate) the structure $\mathbf{\Gamma}$.
6. Check the CP-DAG structure and update the model (both algebraic structure and statical hypothesises)

$$\mathbf{a} \mapsto G|_{\text{CP}} \mapsto f_G(\mathbf{w}).$$

7. Make several steps of the optimization of the model parameters $\mathbf{w}$,

$$\nabla_{\mathbf{w}} = \frac{\partial \mathcal{L}}{\partial \mathbf{w}}.$$

8. Estimate the Bayesian hyperparameters $\mathbf{A}, \mathbf{\Sigma}$ with the optimized model parameters $\mathbf{w}$ and the loss $\mathcal{L}(\mathbf{w})$.
9. Repeat from the step 4.

3 Case for the linear model

For a linear model this algorithm is equivalent to the quadratic optimization problem ^{1, 2}

$$\min_{\mathbf{a}} \mathcal{L}(\mathbf{w}; \mathbf{x}, f_G) = \lambda \mathbf{a}^\top \mathbf{Q} \mathbf{a} + (1 - \lambda) \mathbf{x}^\top \mathbf{a}.$$

4 Covariant lasso as a control of the DAG structure

Multidimensional lasso instead or together with the normal regularization, covariant elastic net. A d -dimensional Laplace distribution with mean vector $\boldsymbol{\mu}$ and positive-definite scale matrix $\mathbf{A}$

$$\mathcal{L}_{\mathbf{w}} = \frac{2}{(2\pi)^{d/2} |\mathbf{A}|^{1/2}} \left(\frac{1}{\sqrt{2}} \sqrt{(\mathbf{w} - \boldsymbol{\mu})^\top \mathbf{A}^{-1} (\mathbf{w} - \boldsymbol{\mu})} \right)^\nu K_\nu \left(\sqrt{2} (\mathbf{w} - \boldsymbol{\mu})^\top \mathbf{A}^{-1} (\mathbf{w} - \boldsymbol{\mu}) \right),$$

where $\nu = d/2 - 1$ and K_ν is the modified Bessel function of the second kind.

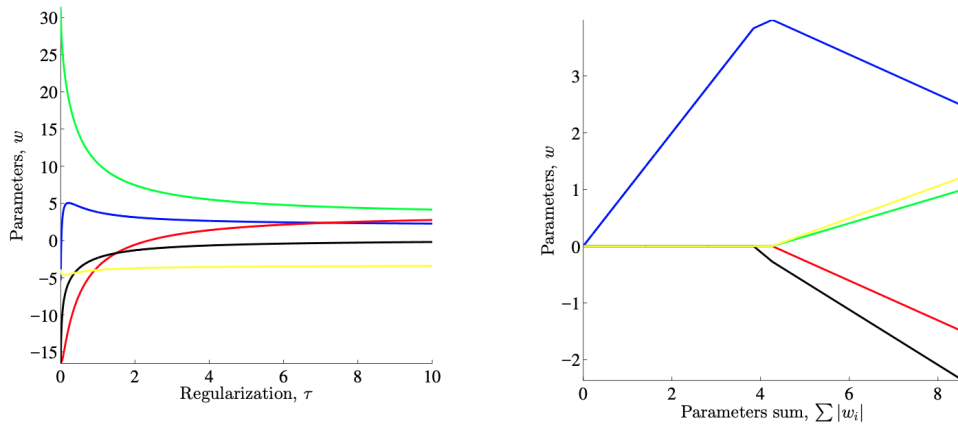


Figure 4: The ridge and lasso regularization for different values of the covariance matrix $\mathbf{A}$.

¹Quadratic programming feature selection for multicorrelated signal decoding with partial least squares <https://doi.org/10.1016/j.eswa.2022.117967>

²Comprehensive study of feature selection methods to solve multicollinearity problem according to evaluation criteria <https://doi.org/10.1016/j.eswa.2017.01.048>